

AEGIS-Dengue: Multi-Horizon GCN-GRU Forecasting for District-Level Dengue Early Warning in Sri Lanka

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Abstract

Dengue early-warning systems are often evaluated only one week ahead and without strong baseline comparisons. In this study, we develop a district-level forecasting system for Sri Lanka using weekly dengue cases from 25 districts, combined with climate data. We evaluate the model using nine walk-forward folds and compare it with persistence and seasonal-naïve baselines. We test several improvements, including learnable climate delays, different graph structures, and spatial-temporal features. At one week ahead, none consistently outperforms persistence. However, at three and four weeks ahead, the model performs better, showing that climate and spatial information become more useful when forecasts are made further in advance.

1 Problem Motivation

Weekly district-level dengue surveillance can support earlier public-health action when forecasts provide useful lead time before a case surge. Short-horizon forecasting is difficult because the most recent case count is already a very strong predictor of the next observation, while longer-horizon forecasts must rely more on learned temporal structure, climate, and other contextual information.

A useful dengue early-warning system should therefore be evaluated not only by average error, but also by its ability to forecast high-incidence periods. This study focuses on district-level forecasts for horizons $h \in \{1, 2, 3, 4\}$ and evaluates both overall MAE and peak MAE under leakage-controlled temporal backtesting.

Moreover, district-level forecasts can support earlier vector-control resource allocation by the National Dengue Control Unit (NDCU) and Public Health Inspectors (PHIs), help hospitals and healthcare providers prepare staff and ward capacity ahead of surges, and enable more targeted intervention in high-burden districts.

2 Identified Datasets and Sources

This study integrates epidemiological surveillance and climate observations for district-level dengue forecasting in Sri Lanka.

Dengue Case Counts: Weekly confirmed dengue case counts were obtained from the **denguedatahub** open research repository [7]. The raw data contain 26 reporting areas, which were consolidated into 25 administrative districts by merging Kalmunai with Ampara. The final panel spans 1,012 reporting periods from December 2006 to May 2026. These weekly district-level case counts form the forecasting target.

Weather Data: Meteorological variables were extracted from ERA5-Land through Google Earth Engine¹ [6]. Variables include

rainfall, 2 m air temperature, dew-point temperature, and eastward and northward wind components. The data were aggregated to the same district-period calendar used for dengue surveillance.

Rainfall Validation: Climate Hazards Group InfraRed Precipitation with Station Data (CHIRPS) [2] was independently extracted through Google Earth Engine² and used to validate rainfall extraction. CHIRPS is used as a validation source rather than as a model input.

3 Forecasting Framework - Initial Baseline

3.1 GCN-GRU Architecture

For each input week, we use the graph convolution [4] combining information from the different districts based on the selected adjacency matrix. The resulting spatial information is then passed through a GRU [1], which captures how the patterns change over the previous weeks. Finally, a linear layer predicts the change in log-transformed dengue cases for the next week. We use an anchored residual target,

$$\Delta_{t,h} = \log(1 + y_{t+h}) - \log(1 + y_t),$$

where a prediction of zero means that the model expects the number of cases to remain at the same level as the current week.

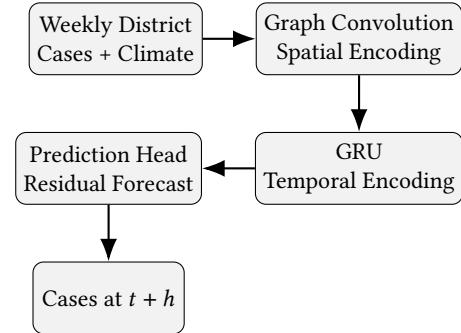


Figure 1: AEGIS-Dengue forecasting pipeline. The adjacency is experimentally varied between identity, geographic contiguity, Gaussian distance, and learned adaptive graphs.

We introduce two feature variants. **v0** contains the dengue cases and climate features, while **v1** adds hand-specified climate delay averages for 4/8/12 weeks. An identity adjacency graph is used as the no-neighbor control; under this setting graph convolution becomes a per-node transformation and the model behaves as the GRU-only control.

¹https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_HOURLY

²https://developers.google.com/earth-engine/datasets/catalog/UCSB-CHG_CHIRPS_DAILY

Table 1: One-week-ahead baseline results over the seven headline folds.

Model	Feat.	MAE	RMSE	Peak MAE	2017 MAE
Persistence	–	16.42	33.61	26.61	36.08
GRU only	v1	16.68	35.25	28.06	42.97
GRU only	v0	16.86	35.94	28.43	43.53
GCN+GRU	v0	18.60	39.68	29.81	51.87
GCN+GRU	v1	18.98	40.36	30.36	54.81

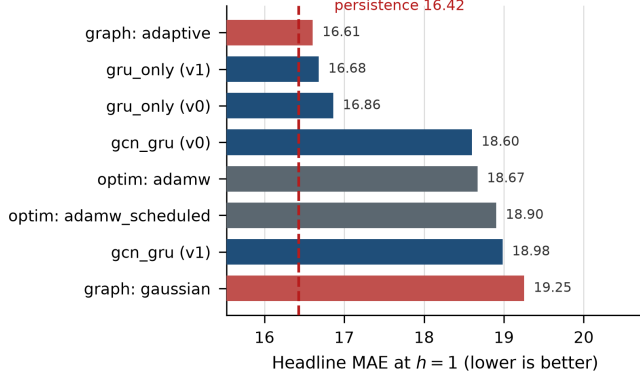


Figure 2: Every one-week configuration against the persistence baseline (dashed). Architecture and feature variants in blue, graph representations in red, and optimizer variants in grey.

3.2 Experimental Setup

Temporal evaluation. All experiments use nine walk-forward folds with expanding training history. The headline mean uses folds 1, 2, 3, 6, 7, 8, and 9. Fold 1 contains the 2017 epidemic and is analyzed separately.

Metrics. The main metrics are MAE, RMSE, and **peak MAE**, where peak cells are determined from each district’s historical distribution. Multiple-seed experiments use three seeds and report the seed standard deviation.

Baselines. Persistence copies the most recent observed case count to the forecast horizon. Seasonal naive uses the corresponding observation from the previous year. Neural comparisons include the geographic GCN-GRU and its identity-graph GRU control.

4 Results and Model Improvement Experiments

4.1 One-Week Baseline Results

Based on the baseline results at one week ahead ($h = 1$), **persistence** remains the strongest reliable baseline (Table 1).

The GRU-only control outperforms the geographic GCN-GRU on every fold. Therefore, the queen-contiguity graph is not neutral: it introduces a consistent accuracy cost. The hand-specified climate lag features also do not provide a robust gain.

This pattern persists across every intervention attempted at one week ahead. Figure 2 places the architecture, graph, and optimizer variants reported throughout Section 4 on a common axis against the persistence baseline.

Table 2: One-week objective-function experiment.

Objective	Headline MAE	2017 MAE	Seed sd
Persistence	16.42	36.08	–
Level-weighted	16.59	38.06	0.57
Quantile	17.14	42.75	0.36
Huber-weighted	17.71	46.66	0.51
Huber	18.17	49.76	0.65
Baseline MSE	18.98	54.81	0.68

4.2 Learnable Climate Lags

To better account for the delayed effect of climate on dengue cases, we added a learnable lag mechanism to the baseline model. Each district learns its own 26-week delay kernel, represented as a softmax mixture of six Gaussian bumps, which reweights the climate history based on the delays that are most useful for forecasting.

Cross-correlation analysis identifies rainfall-to-dengue delays of approximately 5–10 weeks across districts, with a median of eight weeks. A learnable lag encoder can recover planted synthetic delays, but end-to-end training on the real one-week forecasting objective does not recover the measured delays or improve accuracy. In the original single-seed experiment, the learnable-lag model obtains 20.67 MAE compared with 18.36 for fixed windows and 18.07 without lag treatment. The learned and measured delays correlate at only $r = -0.16$.

This result is explained by the strength of the most recent case observation at $h = 1$: removing all climate channels from recent normal folds changes MAE by only about +0.01. Thus, the forecasting loss provides little information from which a climate-delay parameter can be learned.

4.3 Objective and Loss Improvements

The baseline is trained on masked MSE over a log-residual target but evaluated using raw-count MAE. A level-weighted objective was therefore tested together with Huber, weighted Huber, and quantile losses.

Level weighting reduces the GCN-GRU headline MAE from 18.98 to 16.59, but approximately all of this gain comes from the 2017 epidemic fold. Across the other six headline folds the change is essentially zero. The seed-mean ensemble reaches 16.36 MAE versus 16.42 for persistence, which is too small relative to seed variation to claim a reliable win. Peak MAE also remains worse than persistence.

4.4 Hyperparameter Tuning

A 50-trial random search was performed over nine axes: lookback, GCN hidden size, GCN depth, GRU hidden size, GRU depth, dropout, learning rate, batch size, and weight decay. Selection used validation folds only.

The selected configuration was lookback 16, GCN hidden size 32, one GCN layer, GRU hidden size 128, two GRU layers, dropout 0.2, learning rate 3×10^{-4} , batch size 32, and weight decay 10^{-5} . Its test headline MAE is 18.23 versus 18.98 for the hand-picked GCN-GRU baseline. The 0.75-MAE gain is below two seed standard deviations and the tuned model still loses to both persistence and the identity

Table 3: Graph representation at one week ahead.

Graph	MAE	Peak MAE	2017 MAE	Seed sd
Adaptive	16.61	28.08	40.69	0.39
Identity	16.68	28.06	42.97	0.53
Contiguity	18.98	30.36	54.81	0.68
Gaussian distance	19.25	30.89	56.81	0.46

Table 4: Headline MAE by horizon.

Backbone	Arm	$h = 1$	$h = 2$	$h = 3$	$h = 4$
Identity	Shared	16.71	20.04	23.15	25.74
Identity	Separate	17.20	20.22	22.89	25.40
Contiguity	Shared	18.19	22.31	25.83	28.65
Contiguity	Separate	19.48	23.41	27.08	29.93
Persistence	–	16.42	20.16	24.58	28.47

control. The response surface is therefore relatively flat, suggesting that hyperparameter choice is not the principal limitation at $h = 1$.

4.5 Optimizer and Learning-Rate Scheduling

Adam [3], AdamW [5], and AdamW with ReduceLROnPlateau were compared using the same training harness. Headline MAEs are 18.98, 18.67, and 18.90, respectively. Neither AdamW nor the plateau scheduler produces an established accuracy gain.

A useful mechanism result is that in 23 of 27 scheduled runs, the learning-rate reduction occurs only after the best checkpoint has already been selected by early stopping. Therefore, a plateau scheduler and early stopping on the same metric interact weakly under the current patience settings.

4.6 Graph Representation

Four graph representations were evaluated through the same model: identity, queen contiguity, Gaussian centroid-distance weights, and a learned adaptive adjacency [8].

No graph reliably beats the identity control. The Gaussian graph is worse than contiguity, while the adaptive graph improves only two of seven headline folds and its 0.07-MAE average gain over identity is negligible. This rejects the original precondition for building a more complex dual geographic-learned graph at $h = 1$.

A per-district gate that learns how much to weigh the graph versus no graph reached 18.36 MAE was no better than skipping the graph entirely (16.88), and statistically no different from a gate frozen halfway, showing it learned almost nothing useful. We had expected isolated districts to learn to ignore the graph; instead, Jaffna, the most isolated district, showed the strongest preference for keeping it, though the gate barely moved from its starting point, so this reversal shouldn't be over-interpreted.

4.7 Multi-Horizon Forecasting

The most important improvement appears when the forecast horizon is increased. Persistence is rescored separately at each horizon because its copied observation becomes progressively stale.

For the identity backbone with separately trained horizon models, the skill relative to same-horizon persistence is -4.7% , -0.3% , $+6.9\%$, and $+10.8\%$ from $h = 1$ to $h = 4$. Peak-MAE skill follows

Table 5: MAE cost of each climate ablation, paired over the seven headline folds. Positive values indicate that the ablation degraded accuracy.

Horizon	Removed	Shuffled	Shuffled peak	p
$h = 1$	-0.25	$+0.02$	$+0.40$	0.926
$h = 2$	-0.35	-0.06	$+0.38$	0.854
$h = 3$	-0.06	$+0.43$	$+1.65$	0.226
$h = 4$	$+0.76$	$+1.55$	$+3.10$	0.016

the same pattern: -11.7% , -5.1% , $+5.2\%$, and $+9.8\%$. Figure 3 shows both series together with the paired fold-level evidence.

At $h = 4$, the separate model improves MAE on all seven headline folds, with a mean difference of -3.07 MAE, $t = -4.04$, and $p = 0.0068$. The shared multi-horizon model also wins all seven folds at $h = 4$, with $p = 0.0099$. Unlike several one-week improvements, this effect is not driven by the 2017 epidemic: excluding fold 1, the separate model still improves by 2.71 MAE and wins all six remaining headline folds. This is the first broad and statistically supported improvement observed in the project.

The shared model is statistically similar in accuracy to the separate models on the identity backbone but requires roughly one quarter of the trunk training cost, making it attractive when all four horizons are required operationally.

4.8 Climate Ablation at Longer Horizons

The multi-horizon result establishes that longer-horizon forecasting produces skill, but not that climate is the cause. Longer-range case autocorrelation and seasonality are alternative explanations, and skill at $h = 4$ is also consistent with persistence degrading faster than the model does. A three-arm ablation separates these.

Removing climate entirely is confounded: dropping 16 of the 23 v1 channels changes the input width, the first-layer parameter count, and the effective regularization simultaneously, so any resulting penalty need not concern weather. A shuffle control was therefore added that retains all channels but permutes the climate channels in time within each district. This holds width, parameter count, per-fold scaler statistics, and per-district marginal distributions exactly fixed, and destroys only the alignment between weather and time. Permutation is applied to whole windows, within each district and within each data split, so no weather crosses a district or a train-test boundary, and the target is never modified.

The ablation cost rises monotonically with horizon but reaches significance only at $h = 4$ ($p = 0.016$; Table 5).

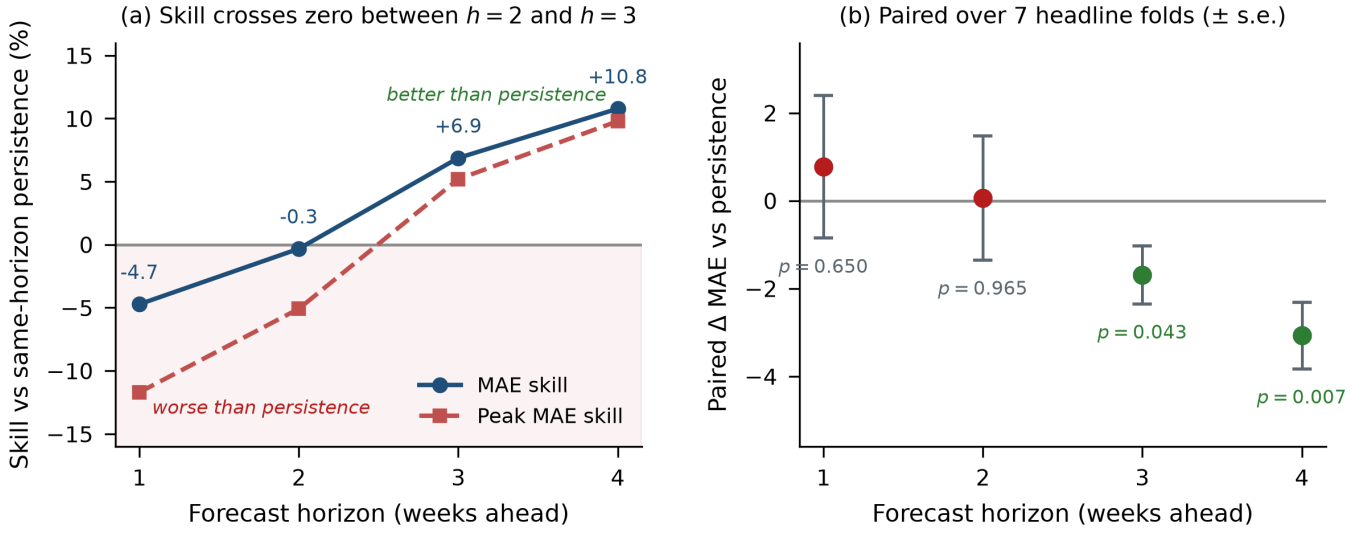


Figure 3: Forecast skill relative to same-horizon persistence, identity backbone with separately trained horizon models. (a) Both MAE and peak-MAE skill cross from negative to positive between $h = 2$ and $h = 3$. (b) The paired per-fold MAE difference over the seven headline folds ($\pm$ standard error); green marks $p < 0.05$. At $h = 4$ the model improves on all seven folds.

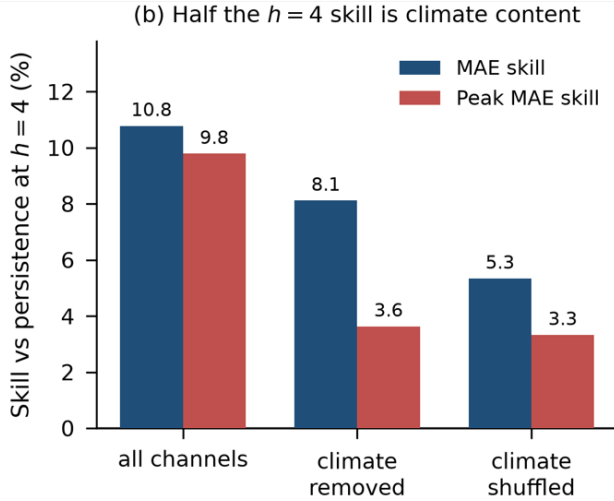


Figure 4: Climate ablation at $h = 1-4$. The MAE cost of removing or shuffling climate rises monotonically with the forecast horizon; the shuffle control holds input width and parameter count fixed, so only the weather-to-time alignment differs.

Table 5 and Figure 4 report the outcome. At $h = 4$ the shuffle control costs +1.55 MAE ($p = 0.016$, 2.4 seed standard deviations) and +3.10 peak MAE ($p = 0.023$), removing approximately half of the MAE skill reported in Table 4 and about two thirds of the peak-MAE skill. Both costs rise monotonically with the horizon, which is the pattern implied by the measured 5–10 week rainfall-to-dengue delay and not the pattern that persistence decay alone would produce. The effect is not concentrated in the epidemic fold: the shuffled arm degrades accuracy on six of seven folds, and

excluding fold 1 the cost is essentially unchanged at +1.27 MAE. At $h = 1$ the cost on recent normal folds is +0.17 MAE, consistent with the near-zero value measured in the learnable-lag analysis, which serves as a correctness check on the ablation harness.

Two limits are worth stating. Climate does not account for all of the gain, since the shuffled arm still outperforms persistence by 5.3% at $h = 4$; persistence decay is therefore a genuine component. In addition, $h = 3$ does not reach significance on its own ($p = 0.226$). Notably, the removal arm alone is weak (+0.76, $p = 0.267$) and negative at $h = 1-3$, so an ablation conducted without the shuffle control would have supported the opposite conclusion.

5 Conclusion

AEGIS-Dengue provides a reproducible, leakage-controlled evaluation of GCN-GRU dengue forecasting across 25 Sri Lankan districts. The full experiments overturn the preliminary conclusion that the one-week neural models outperform persistence: across nine walk-forward folds and three seeds, persistence remains the strongest reliable one-week baseline, and geographic graph convolution consistently hurts relative to the identity control.

The main positive result is multi-horizon forecasting. At three and four weeks ahead, the identity-backbone model reliably outperforms same-horizon persistence on both overall and peak MAE, with the four-week improvement occurring across all seven headline folds at $p < 0.01$. A capacity-matched shuffle ablation attributes roughly half of this four-week skill to climate content, with a cost that grows monotonically with the forecast horizon; the remainder is attributable to the decay of the persistence baseline. Layer normalization further improves seed stability even though it does not establish an accuracy gain. Future work should evaluate fixed measured climate lags, count-based likelihoods, and calibrated multi-quantile forecasts at horizons where the model has demonstrated real predictive skill.

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